

Assignment 6: Data Scraping

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on data scraping.

Directions

1. Rename this file `<FirstLast>_A10_DataScraping.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:
 - Load the packages `tidyverse`, `rvest`, and any others you end up using.
 - Check your working directory

```
#1 setting up packages
library(tidyverse)
library(rvest)
library(lubridate)
library(here)

getwd()
```

```
## [1] "/home/guest/EDE_Spring2026"
```

2. We will be scraping data from the NC DEQs Local Water Supply Planning website, specifically the Durham’s 2024 Municipal Local Water Supply Plan (LWSP):
 - Navigate to <https://www.ncwater.org/WUDC/app/LWSP/search.php>
 - Set the year to 2024
 - Scroll down and select the LWSP link next to Durham Municipality.
 - Note the web address: <https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024>

Indicate this website as the as the URL to be scraped. (In other words, read the contents into an `rvest` webpage object.)

```
#2
#scraping website
the_url <-
"https://www.ncwater.org/WUDC/app/LWSP/report.php?psid=03-32-010&year=2024"
the_website <- read_html(the_url)
```

3. The data we want to collect are listed below:

- From the “1. System Information” section:
- Water system name
- Contact Person
- Ownership
- From the “3. Water Supply Sources” section:
- Maximum Day Use (MGD) - for each month

In the code chunk below scrape these values, assigning them to four separate variables.

HINT: The first value should be “Durham”, the second “Reginald Hicks”, the third “Municipality”, and the last should be a vector of 12 numeric values (represented as strings)“.

```
#3
#reading all tables on the page into a list
all_tables <- the_website %>%
  html_nodes("table") %>%
  html_table(fill = TRUE)

#water system name
water_system_name <- the_website %>%
  html_nodes("table:nth-of-type(1) tr:nth-child(1) td:nth-child(2)") %>%
  html_text() %>%
  .[1]
water_system_name
```

```
## [1] "Durham"
```

```
#contact person
contact_person <- the_website %>%
  html_nodes("table:nth-of-type(1) tr:nth-child(3) td:nth-child(2)") %>%
  html_text() %>%
  .[1]
contact_person
```

```
## [1] "2-24"
```

```
#ownership
ownership <- the_website %>%
  html_nodes("table:nth-of-type(1) tr:nth-child(2) td:nth-child(4)") %>%
  html_text() %>%
  .[1]
ownership
```

```
## [1] "Municipality"
```

```
#MGD for each month
monthly_table <- all_tables[[13]]

max_withdrawals <- c(
  monthly_table[[3]][1:4], # Jan, Feb, Mar, Apr
  monthly_table[[6]][1:4], # May, Jun, Jul, Aug
  monthly_table[[9]][1:4]  # Sep, Oct, Nov, Dec
)

max_withdrawals
```

```
## [1] 34.50 32.10 38.20 36.73 36.06 46.65 41.90 42.96 37.33 37.36 36.58 34.45
```

```
length(max_withdrawals)
```

```
## [1] 12
```

4. Convert your scraped data into a dataframe. This dataframe should have a column for each of the 4 variables scraped and a row for the month corresponding to the withdrawal data. Also add a Date column that includes your month and year in data format. (Feel free to add a Year column too, if you wish.)

TIP: Use `rep()` to repeat a value when creating a dataframe.

NOTE: It's likely you won't be able to scrape the monthly withdrawal data in chronological order. You can overcome this by creating a month column manually assigning values in the order the data are scraped: "Jan", "May", "Sept", "Feb", etc...

5. Create a line plot of the maximum daily withdrawals across the months for 2024, making sure, the months are presented in proper sequence.

```
#4

#month order according to scraping
month_order <- c(1,2,3,4, 5,6,7,8, 9,10,11,12)

df_withdrawals <- data.frame(
  "Month"      = month_order,
  "Year"       = rep(2024, 12),
  "Max_Use_mgd" = as.numeric(max_withdrawals),
  "Water_System" = rep(water_system_name, 12),
```

```

"Contact_Person" = rep(contact_person, 12),
"Ownership"      = rep(ownership, 12)
) %>%
  mutate(Date = my(paste(Month, "-", Year)))

glimpse(df_withdrawals)

```

```

## Rows: 12
## Columns: 7
## $ Month      <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12
## $ Year       <dbl> 2024, 2024, 2024, 2024, 2024, 2024, 2024, 2024, 2024, 2024, 2024, 2024
## $ Max_Use_mgd <dbl> 34.50, 32.10, 38.20, 36.73, 36.06, 46.65, 41.90, 42.96, 42.96, 42.96, 42.96, 42.96
## $ Water_System <chr> "Durham", "Durham", "Durham", "Durham", "Durham", "Durham", "Durham", "Durham", "Durham", "Durham", "Durham", "Durham"
## $ Contact_Person <chr> "2-24", "2-24", "2-24", "2-24", "2-24", "2-24", "2-24", "2-24", "2-24", "2-24", "2-24", "2-24"
## $ Ownership    <chr> "Municipality", "Municipality", "Municipality", "Municipality", "Municipality", "Municipality", "Municipality", "Municipality", "Municipality", "Municipality", "Municipality", "Municipality"
## $ Date        <date> 2024-01-01, 2024-02-01, 2024-03-01, 2024-04-01, 2024-05-01, 2024-06-01, 2024-07-01, 2024-08-01, 2024-09-01, 2024-10-01, 2024-11-01, 2024-12-01

```

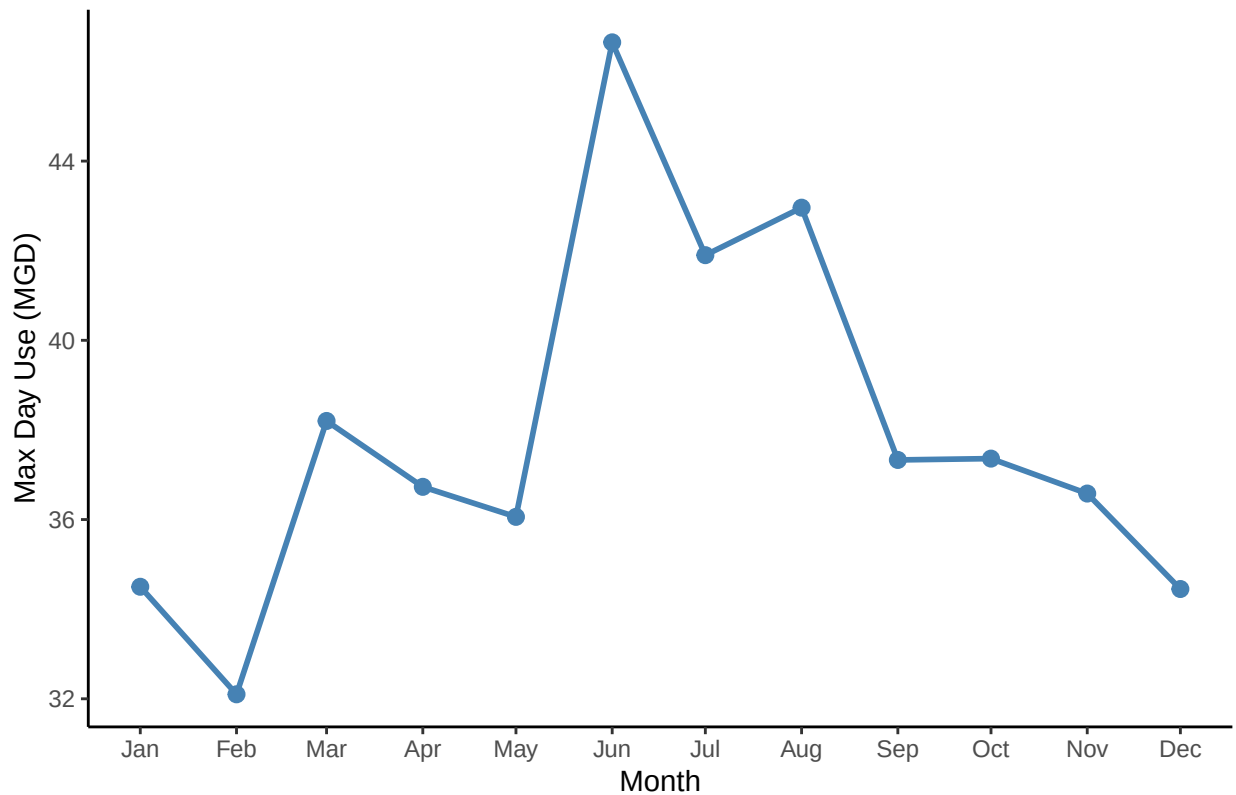
#5

```

#creating a line plot
ggplot(df_withdrawals, aes(x = Date, y = Max_Use_mgd)) +
  geom_line(color = "steelblue", linewidth = 1) +
  geom_point(color = "steelblue", size = 2.5) +
  scale_x_date(date_labels = "%b", date_breaks = "1 month") +
  labs(
    title = "2024 Max Daily Water Withdrawals - Durham",
    x      = "Month",
    y      = "Max Day Use (MGD)"
  ) +
  theme_classic()

```

2024 Max Daily Water Withdrawals – Durham



6. Note that the PWSID and the year appear in the web address for the page we scraped. Construct a function with two inputs, “PWSID” and “year”, that:

- Creates a URL pointing to the LWSP for that PWSID for the given year
- Creates a website object and scrapes the data from that object (just as you did above)
- Constructs a dataframe from the scraped data, mostly as you did above, but includes the PWSID and year provided as function inputs in the dataframe.
- Returns the dataframe as the function’s output

#6.

#constructing func

```
scrape.it <- function(PWSID, year) {
```

#creating URL from inputs

```
the_url <- paste0(
  "https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=",
  PWSID, "&year=", year
)
```

#fetching page

```
the_website <- read_html(the_url)
```

#reading all tables

```
all_tables <- the_website %>%
```

```

html_nodes("table") %>%
html_table(fill = TRUE)

#scraping system info
water_system_name <- the_website %>%
  html_nodes("table:nth-of-type(1) tr:nth-child(1) td:nth-child(2)") %>%
  html_text() %>%
  .[1]

contact_person <- the_website %>%
  html_nodes("table:nth-of-type(1) tr:nth-child(4) td:nth-child(2)") %>%
  html_text() %>%
  .[1]

ownership <- the_website %>%
  html_nodes("table:nth-of-type(1) tr:nth-child(2) td:nth-child(4)") %>%
  html_text() %>%
  .[1]

#finding monthly table
monthly_index <- which(sapply(all_tables, function(t)
  ncol(t) == 9 & nrow(t) == 4))[1]
monthly_table <- all_tables[[monthly_index]]

#scraping max dat use
max_withdrawals <- c(
  monthly_table[[3]][1:4], # Jan, Feb, Mar, Apr
  monthly_table[[6]][1:4], # May, Jun, Jul, Aug
  monthly_table[[9]][1:4]  # Sep, Oct, Nov, Dec
)

#building dataframe
df <- data.frame(
  "Month" = 1:12,
  "Year" = rep(year, 12),
  "Max_Use_mgd" = as.numeric(max_withdrawals),
  "Water_System" = rep(water_system_name, 12),
  "Contact_Person" = rep(contact_person, 12),
  "Ownership" = rep(ownership, 12),
  "PWSID" = rep(PWSID, 12)
) %>%
  mutate(Date = my(paste(Month, "-", Year)))

return(df)
}

```

7. Use the function above to extract and plot max daily withdrawals for Durham (PWSID='03-32-010') for each month in 2020. Then show the structure of the dataframe with either the `str()` or the `glimpse()` function.

```
#7
```

```

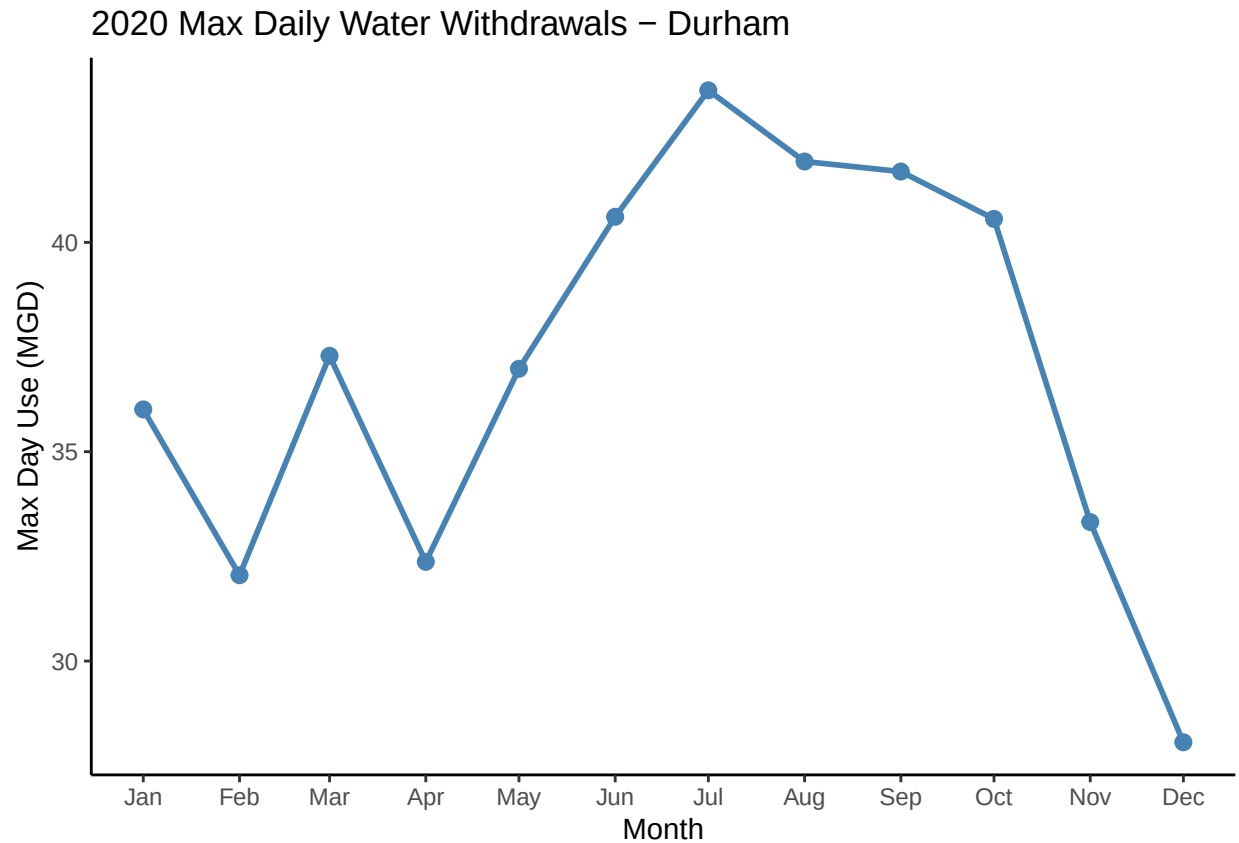
#extracting Durham 2020 data using the function
df_durham_2020 <- scrape.it(PWSID = "03-32-010", year = 2020)

#showing structure
glimpse(df_durham_2020)

## Rows: 12
## Columns: 8
## $ Month      <int> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12
## $ Year       <dbl> 2020, 2020, 2020, 2020, 2020, 2020, 2020, 2020, 2020, 2020, 2020, 2~
## $ Max_Use_mgd <dbl> 36.01, 32.05, 37.29, 32.37, 36.98, 40.61, 43.63, 41.93, ~
## $ Water_System <chr> "Durham", "Durham", "Durham", "Durham", "Durham", "Durh~
## $ Contact_Person <chr> "Reginald Hicks", "Reginald Hicks", "Reginald Hicks", "~
## $ Ownership   <chr> "Municipality", "Municipality", "Municipality", "Munici~
## $ PWSID       <chr> "03-32-010", "03-32-010", "03-32-010", "03-32-010", "03~
## $ Date        <date> 2020-01-01, 2020-02-01, 2020-03-01, 2020-04-01, 2020-05~

#plotting
ggplot(df_durham_2020, aes(x = Date, y = Max_Use_mgd)) +
  geom_line(color = "steelblue", linewidth = 1) +
  geom_point(color = "steelblue", size = 2.5) +
  scale_x_date(date_labels = "%b", date_breaks = "1 month") +
  labs(
    title = "2020 Max Daily Water Withdrawals - Durham",
    x      = "Month",
    y      = "Max Day Use (MGD)"
  ) +
  theme_classic()

```



8. Use the function above to extract data for Cary (PWSID = '03-92-020') in 2020. Combine this data with the Durham data collected above and create a plot that compares Cary's to Durham's water withdrawals.

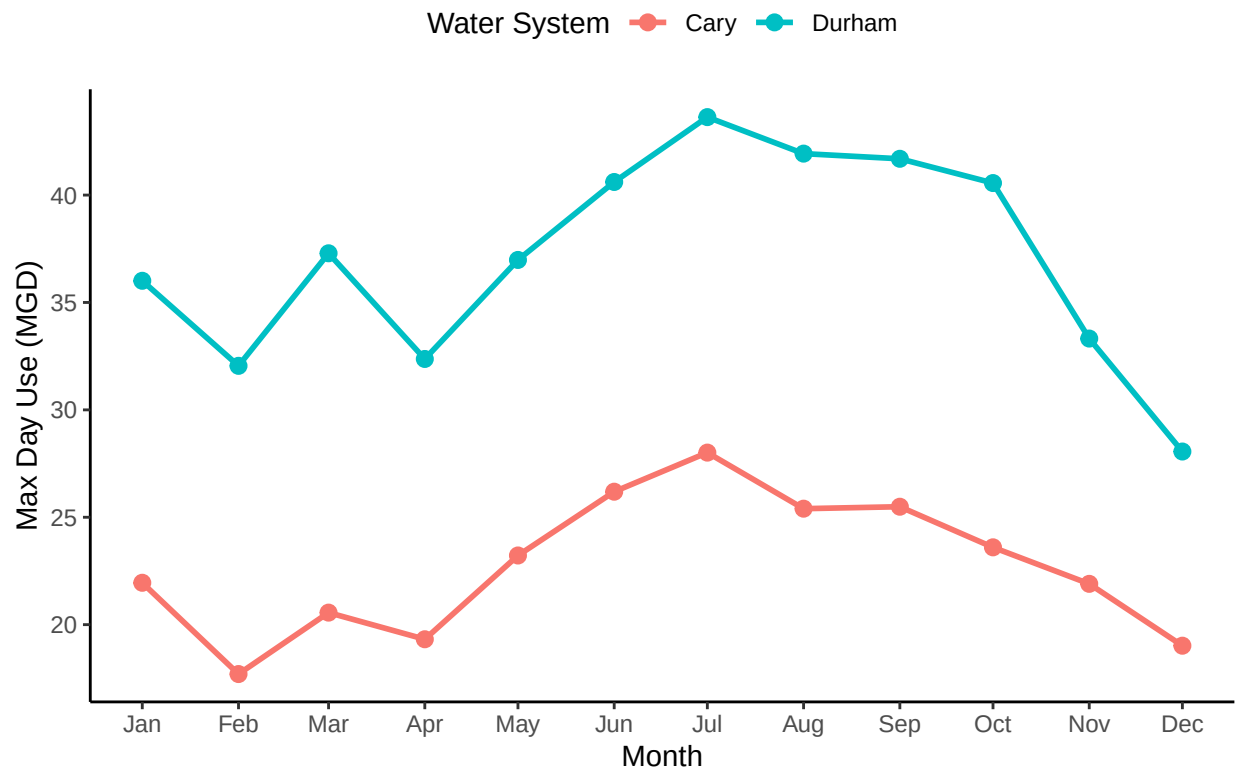
```
#8

#extracting Cary 2020 data
df_cary_2020 <- scrape.it(PWSID = "03-92-020", year = 2020)

#combining Durham and Cary into one dataframe
df_combined_2020 <- bind_rows(df_durham_2020, df_cary_2020)

#comparison plot
ggplot(df_combined_2020, aes(x = Date, y = Max_Use_mgd, color = Water_System)) +
  geom_line(linewidth = 1) +
  geom_point(size = 2.5) +
  scale_x_date(date_labels = "%b", date_breaks = "1 month") +
  labs(
    title = "2020 Max Daily Water Withdrawals: Durham vs. Cary",
    x      = "Month",
    y      = "Max Day Use (MGD)",
    color  = "Water System"
  ) +
  theme_classic() +
  theme(legend.position = "top")
```


2020 Max Daily Water Withdrawals: Durham vs. Cary



9. Use the code & function you created above to plot Cary's max daily withdrawal by months for the years 2018 thru 2023. Add a smoothed line to the plot (method = 'loess').

TIP: See Section 3.2 in the "06_Data_Scraping.Rmd" where we apply "map2()" to iteratively run a function over two inputs (where one of the two inputs does not change). Pipe the output of the map2() function to bind_rows() to combine the dataframes into a single one, and use that to construct your plot.

```
#9

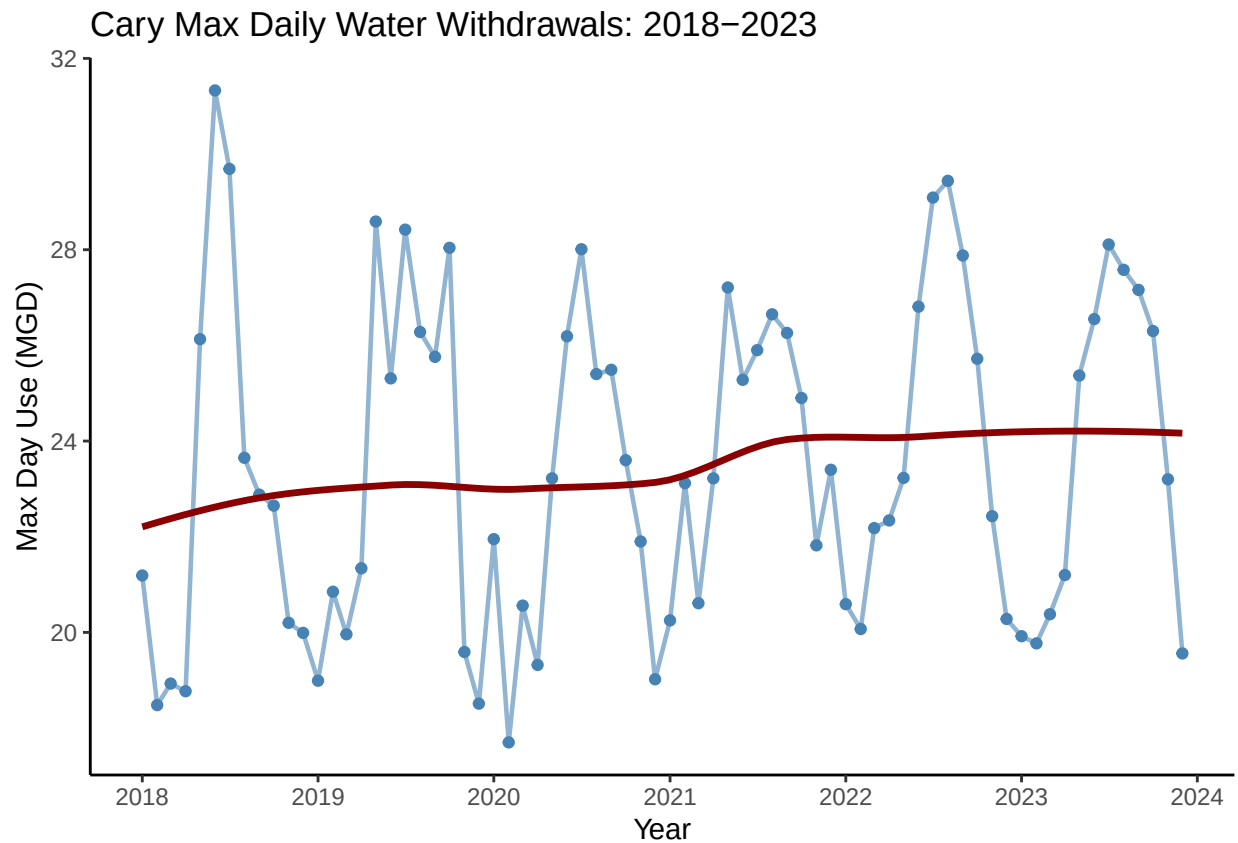
#creating vectors of inputs
the_years <- 2018:2023
cary_pwsids <- rep("03-92-020", length(the_years))

#using map2
df_cary_multiyear <- map2(cary_pwsids, the_years, scrape.it) %>%
  bind_rows()

#plotting w loess smooth
ggplot(df_cary_multiyear, aes(x = Date, y = Max_Use_mgd)) +
  geom_line(color = "steelblue", linewidth = 0.8, alpha = 0.6) +
  geom_point(color = "steelblue", size = 1.5) +
  geom_smooth(method = "loess", se = FALSE,
             color = "darkred", linewidth = 1.2) +
  scale_x_date(date_labels = "%Y", date_breaks = "1 year") +
```

```
labs(
  title = "Cary Max Daily Water Withdrawals: 2018-2023",
  x     = "Year",
  y     = "Max Day Use (MGD)"
) +
theme_classic()
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



Question: Just by looking at the plot (i.e. not running statistics), does Cary have a trend in water usage over time? > Answer: Yes. The LOESS line rises gradually from ~22 MGD in 2018 to ~24 MGD by 2022, then levels off, suggesting modest growth in water demand that appears to be stabilizing in recent years. >